A special mentoring program for master's students in astrophysics promoting DEI

Judith Ardèvol, Toni Bertólez, **Mar Carretero Castrillo***, Lucía Castells Tiestos, Ana Climent Catalá, Roberto Emparan, Óscar Jiménez Arranz, Elina Kefala, Quim Llorens Giralt, Antonia Micol Frassino, Edgar Molina, Marija Tomašević

ICCUB | Institut de Ciències del Cosmos UNIVERSITAT DE BARCELONA | EXCELENCIA MARÍA DE MAEZTU 2020-2024

- DEIC ICCUB -

UNIVERSITAT DE BARCELONA

Outline

- Introduction
  - Context
  - The DEIC-ICCUB Mentoring Program
  - Goals and benefits of mentoring
- Special case: female group in 2022-2023
  - The mentoring program trajectory
  - The female group in 2022-2023
  - A successful mentoring program
- Conclusions and recommendations



DEIC at ICCUB



Julia Peral Calbó
Gender Officer



Esther Pallarés
Administrative



Assumpta Parreño
Chair



Francesca Figueras
Ombudsperson



Àngels Ramos
Ombudsperson



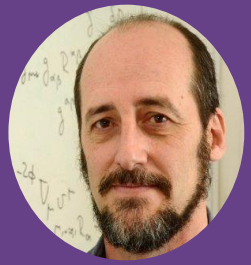
Toni Bertólez
PhD student



Mar Carretero Castrillo
PhD student



Licia Verde
ICREA



Roberto Emparan
ICREA

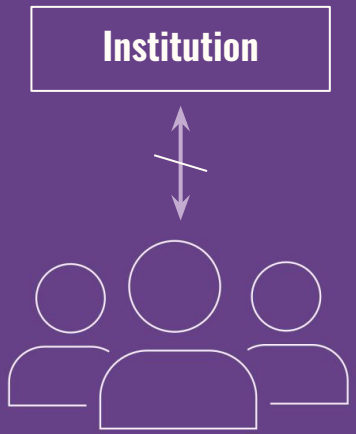


Antonia Frassino
Postdoctoral

Context

Master in Astrophysics, Particle Physics and Cosmology
UB - ICCUB

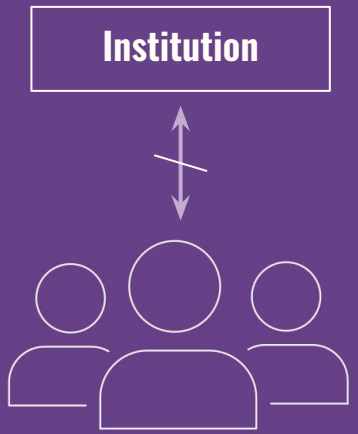
Feeling of disconnection



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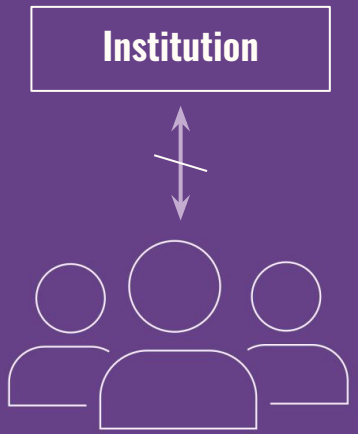
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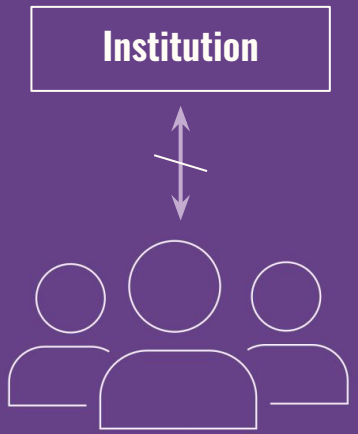
Opportunity to retain people



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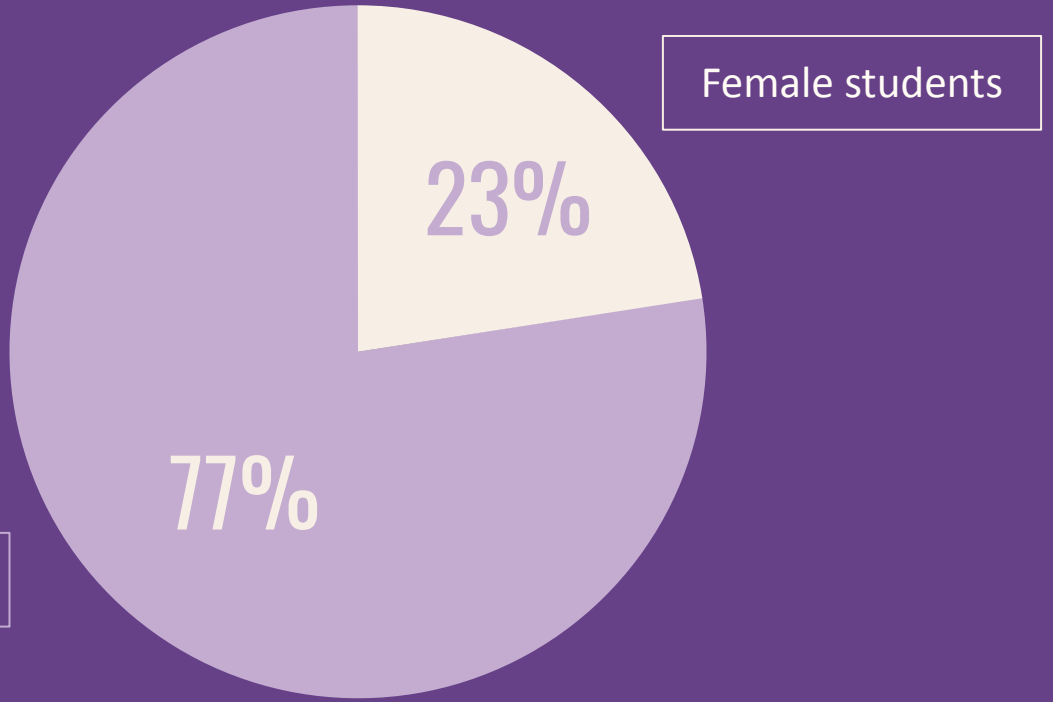


Poor diversity distribution



Poor diversity distribution

From 2020 to 2024:



“**A mentoring program** is a structured initiative where experienced individuals (mentors) guide and support less experienced individuals (mentees) to facilitate their personal and professional growth”

Benefits of mentoring

For mentees

- Guidance and career advice
- Encouragement and self-confidence
- Local network establishment
- Safe and trustworthy space

For mentors

- Exposure to new knowledge and skills
- Satisfaction that you can help students
- Leadership and team management skills
- Combine and refresh your network with a younger one

The DEIC-ICCUB Mentoring Program

Format

- Group mentoring sessions
- Multiple mentors ∼5
- Monthly frequency, 1 h
- Two different groups considered

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Students that feel to belong to minority groups in academia

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- Depend on the group
- DEI topics: share personal experiences, support, and safe space
- Academic topics: master-related, how to approach science (fellowships, …), PhD life

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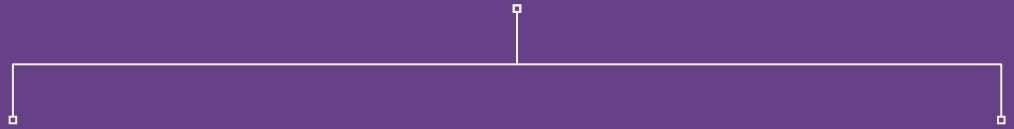
Engagement

- Adaptation of the topics to the students' interests
- Interaction is encouraged
- Some meetings are held in informal spaces (bar, …)

The DEIC-ICCUB Mentoring Program

The role of mentors

TWO KEY ASPECTS



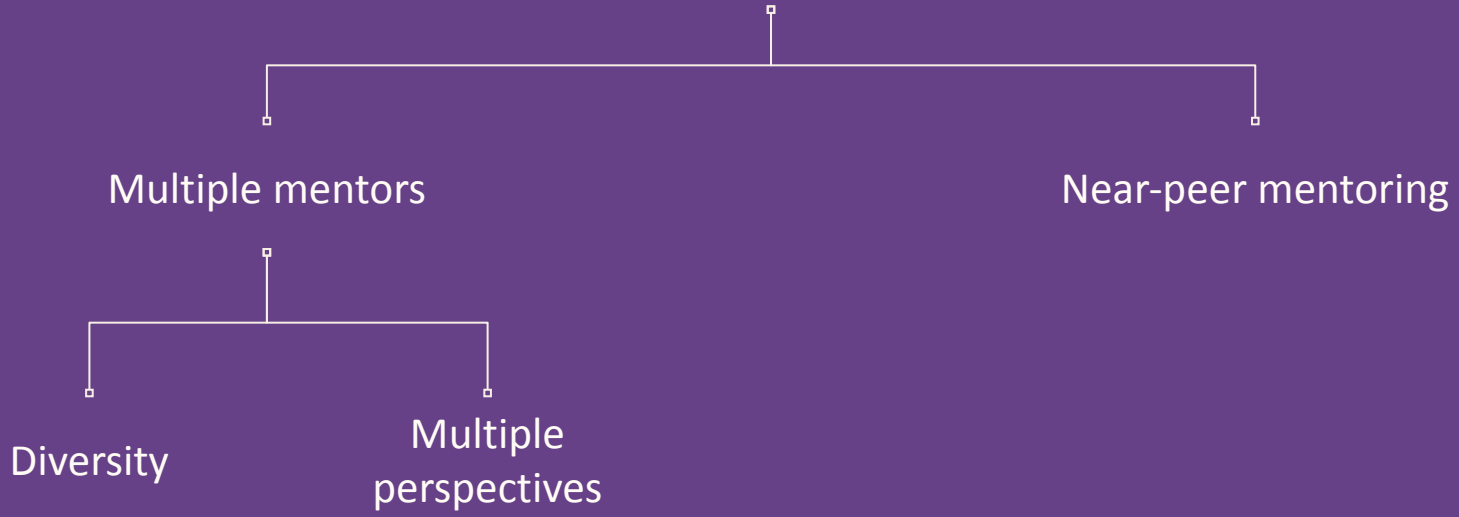
Multiple mentors

Near-peer mentoring

The DEIC-ICCUB Mentoring Program

The role of mentors

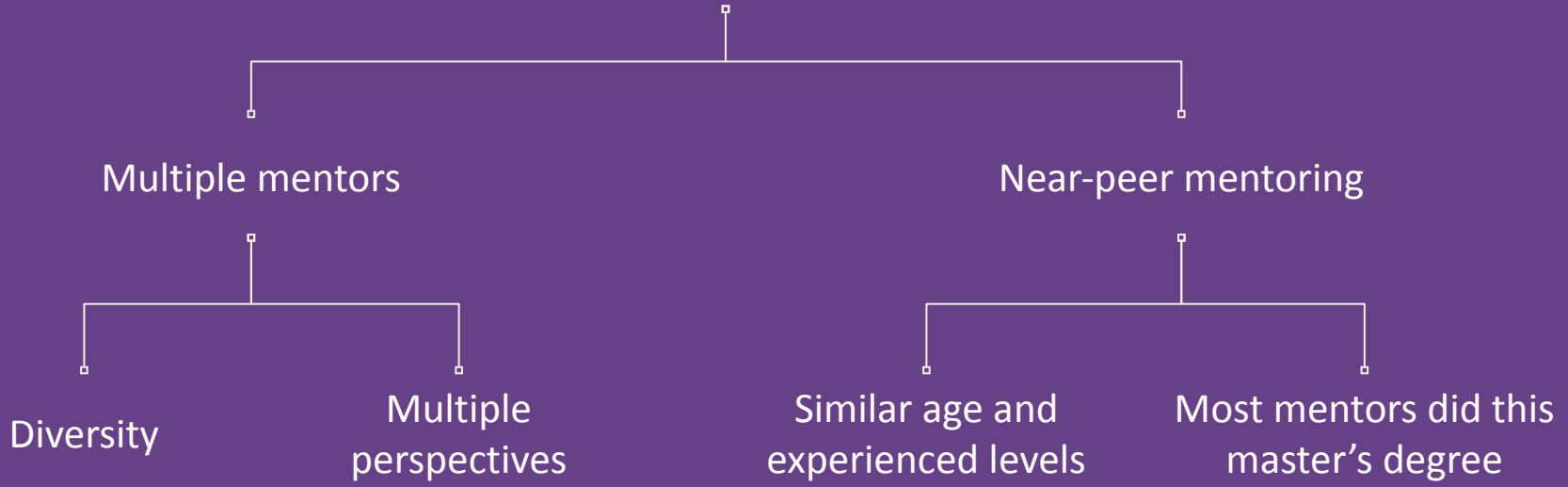
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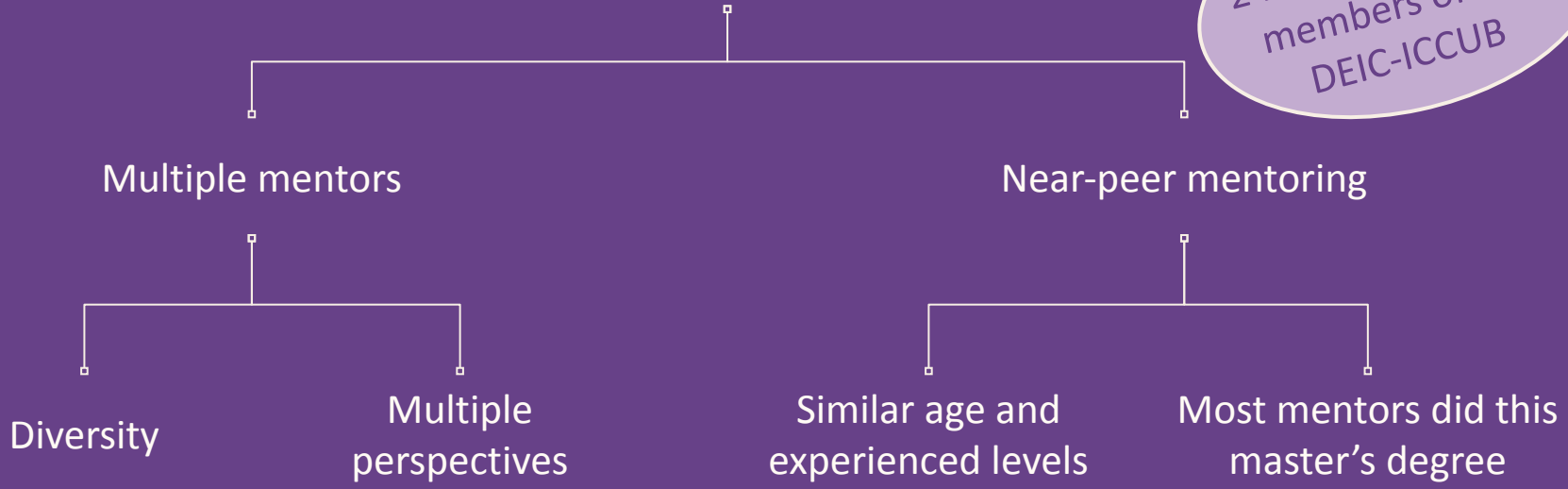


The DEIC-ICCUB Mentoring Program

The role of mentors

TWO KEY ASPECTS

2 PhD mentors are members of the DEIC-ICCUB



The mentoring program trajectory

Start of the program

- Fully remote
- Small version: 4 female mentees
- Mentors: 2 female postdocs
- Only female group

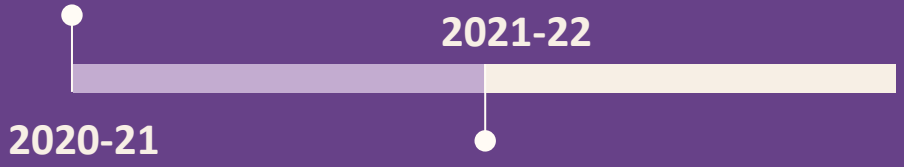


2020-21

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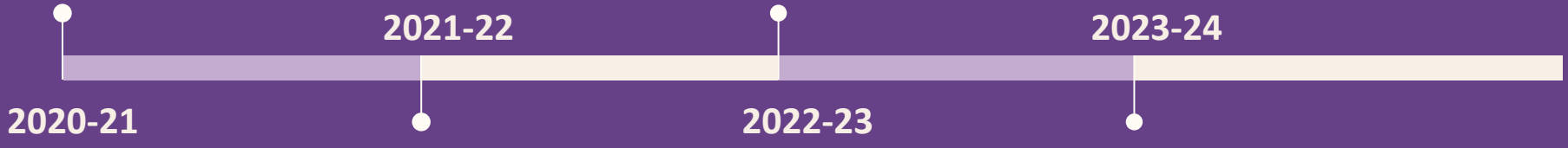
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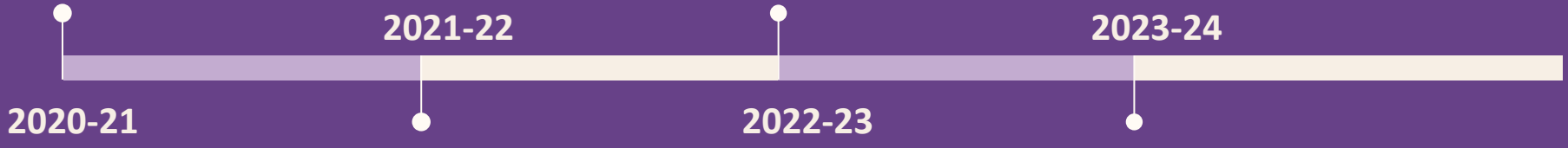
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The female group in 2022-23

Mentors



Mar Carretero Castrillo

PhD student at ICCUB in High Energy Astrophysics



Ana Climent Catalá

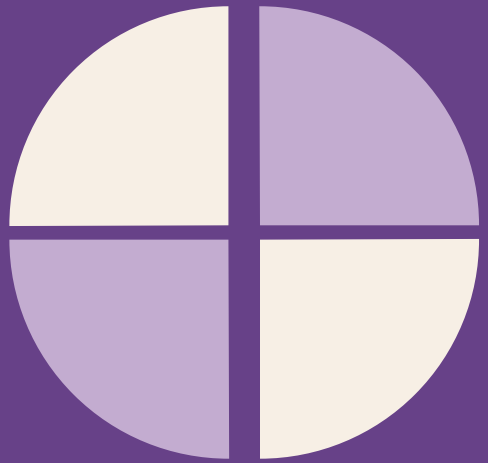
PhD student at ICCUB in High Energy Theory

Mentees

2 master's students in cosmology + 1 master's student in astronomy + 1 master's student in particle physics

The female group in 2022-23

Goals

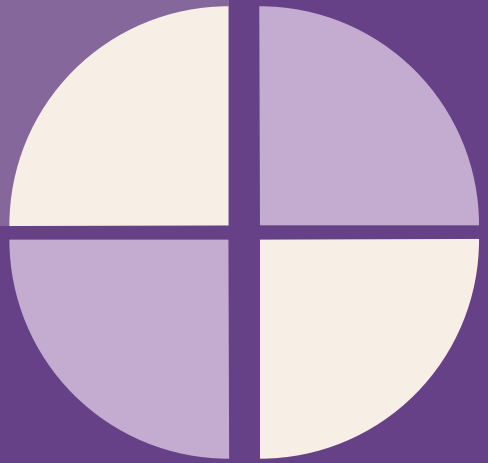


The female group in 2022-23

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BE A SUPPORTIVE GROUP

Group that provide mutual encouragement and assistance



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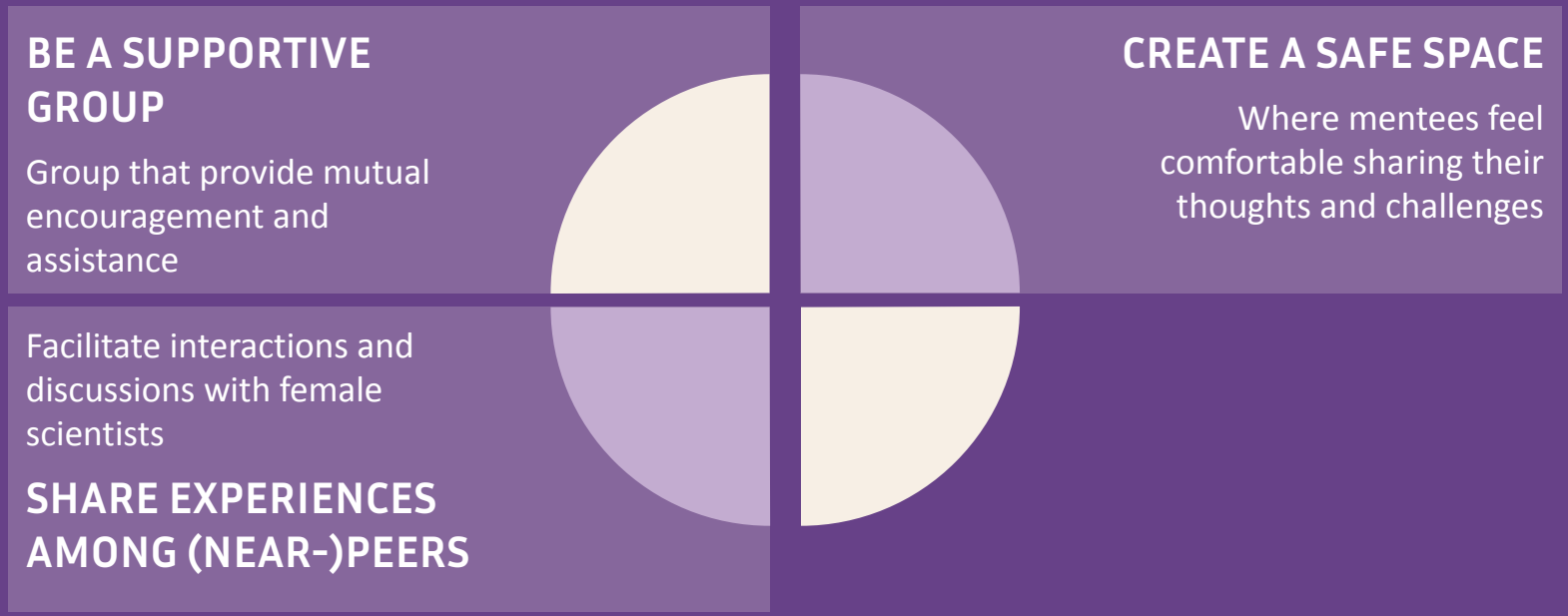


CREATE A SAFE SPACE

Where mentees feel comfortable sharing their thoughts and challenges

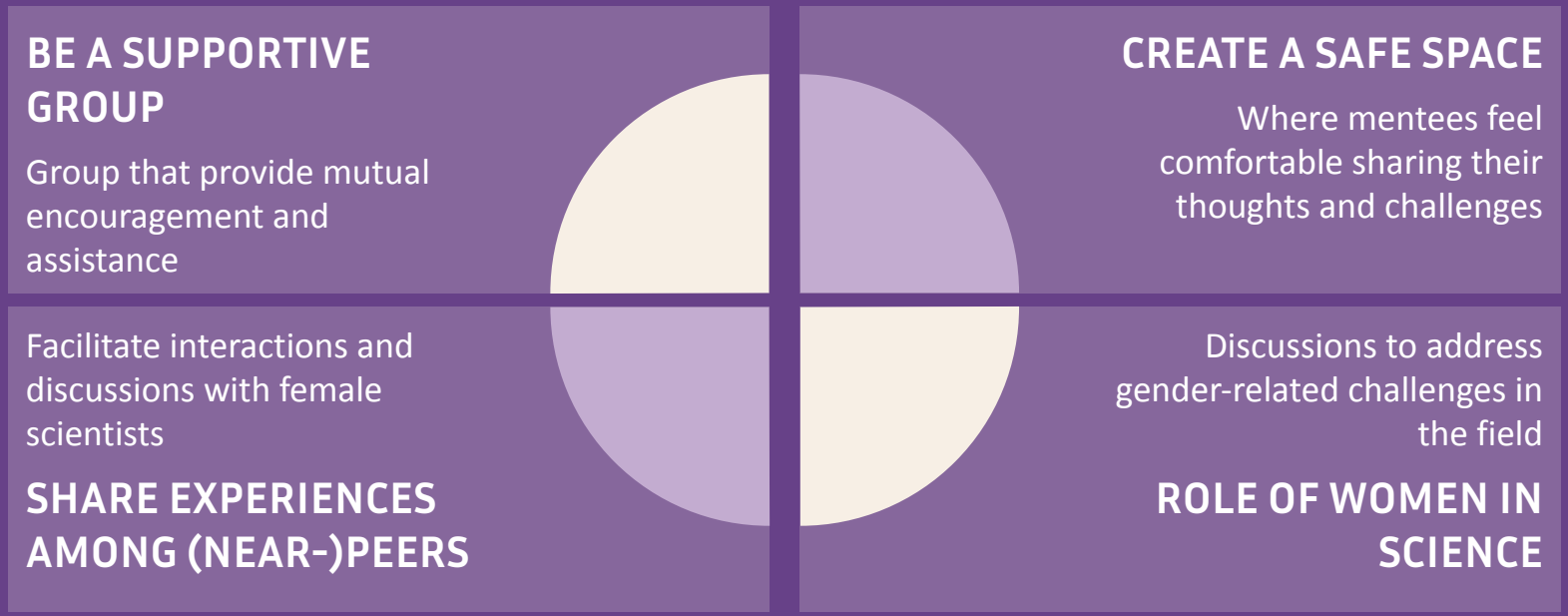
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Goals



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Sessions

Presentations on a topic of interest (by mentors)

+

Group discussions

The female group in 2022-23

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Presentations on a topic of interest (by mentors)

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Topics

- Get familiar among the group:
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  - Difficulties encountered along the way

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  - Finishing master thesis
  - Looking for PhD grants

The female group in 2022-23

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Presentations on a topic of interest (by mentors) + Group discussions

Topics

*Note: mentors did not receive training, we based the sessions on our personal experiences, empathy and active listening

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The female group in 2022-23

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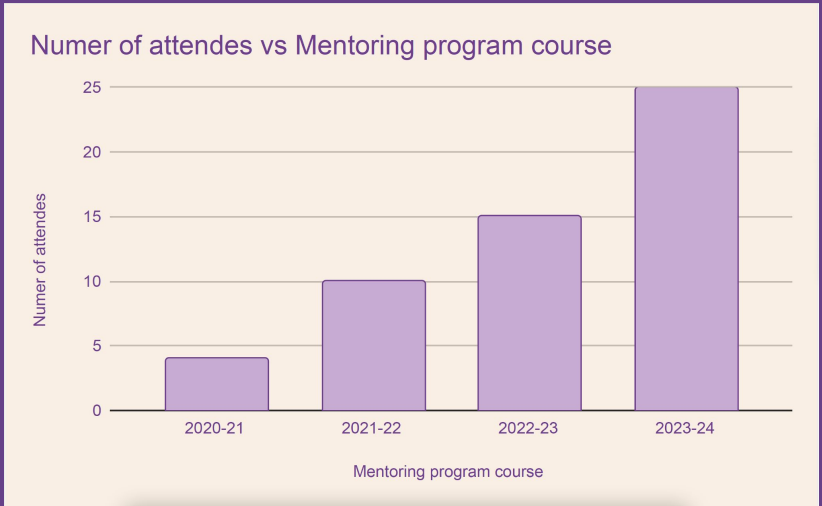
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- Our shared experience as women and connection to the institution were crucial in understanding and addressing specific problems mentees faced (e. g., lack of email responses)

Why is this a successful mentoring program?

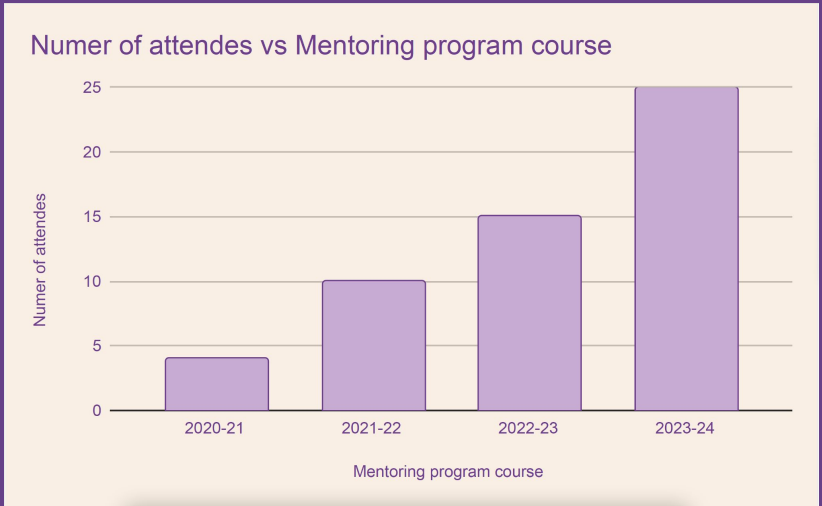
A successful mentoring program

- Effective identification and resolution of conflicts
- Students willing to stay with a PhD at ICCUB
- Students felt guided and supported
- Greater sense of belonging to the institute



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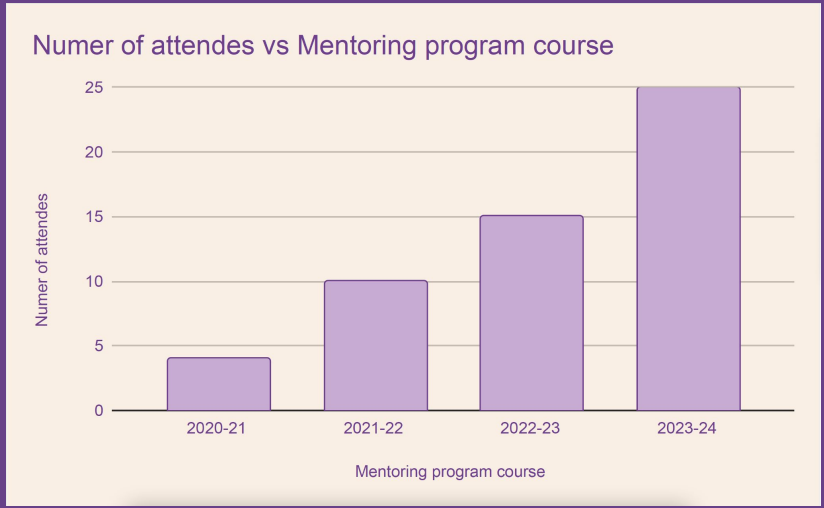
Female group

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2021-22	4/4	2/4
2022-23	2/4*	2/4

*Note: all of them wanted to stay with a PhD at ICCUB

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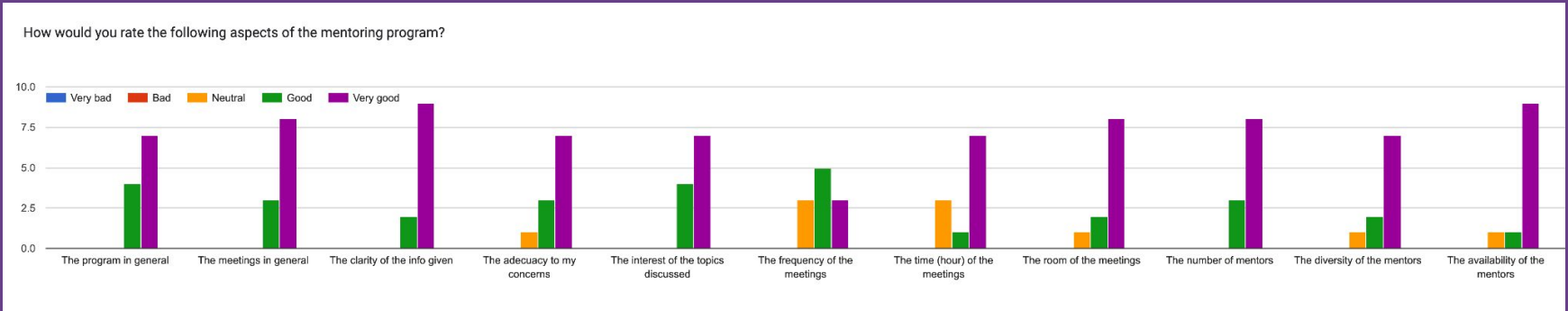
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- High percentage of female mentees chose to pursue a PhD
- *Success:* mentees feel supported, encouraged and confident to enter science, overcoming both external biases in physics and internal obstacles

A successful mentoring program

Reviews



What did you like best of the program?

I think it is a really good idea to help master's students. I really appreciated the information of how one normally starts a PhD and all the links provided!

the open availability of the mentors and the fruitful information given in the meetings

The proximity of the mentors. The chill environment. Amd mostly, the useful advice. Thks

molt utiiil i està guai coneixer PhDs per tenir refs :)

What will you improve?

It would be nice to bring some of the professors that offer PhD grants (in the ICCUB) so they can talk a little bit of what they expect of a PhD student, the difficulties of their own project, etc. Of course this option depends on the disponibility of the professors .

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Literalment res

No haber tenido una versión de este programa durante el grado en física.

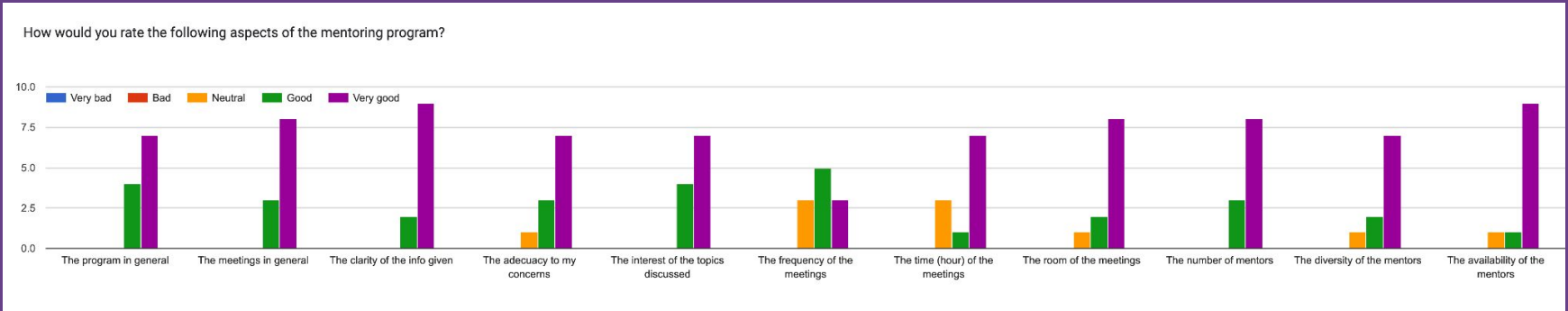
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Conclusions

- Highly useful for students and young researchers, **specially in the case of minorities**
- M**entors** are a **key ingredient** for a successful mentoring
- **Mentors** shape the **perception** of the **institute** and the **PhD life**

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Challenges

- **Limited diversity** among our **mentors** (and **mentees**)
- **Highly dependent** on the **goodwill** and altruism of mentors
- Major challenge: **generational relay of mentors** and **transfer of knowledge**
- **Role of women** in the organization of the program

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Recommendations

- **Training** for mentors? (double-edged sword)
- **Document** all steps and materials to **guarantee the continuity** of the program
- **Institutional support, official recognition,** and DEI commission involved

Do you want to be a mentor (UB people)?
Do you want to apply this to your own institution?

Approach us :)

Thanks

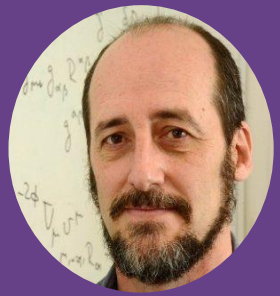






Both mentees and **mentors benefit** from a mentoring program




Backup

Course	Female students	Male students
2020-21	15	34
2021-22	6	20
2022-23	8	31
2023-24	9	38

Backup

Course	Female PhD entering at ICCUB	Male PhD entering at ICCUB
2020-21	1	8
2021-22	2	13
2022-23	4	15
2023-24	11	23